

Developing a Machine Learning Project – Challenges and Checklist

COMP6246: Machine Learning Technologies (MSc)

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Lecture Outline

- By the end of this lecture, you will be able to
 - Gain an overview of a practical workflow for an end-to-end Machine Learning project
 - Understand how to create the steps/checklist to set up an ML project
 - Understand the challenges in Machine Learning and learn how to navigate them.

Challenges in Machine Learning

- ML = data + algorithm → what can go wrong?
- Issues with data
 - Insufficient training data
 - Non-representative data – *sampling bias*
 - Poor quality data – outliers, missing values
 - Irrelevant features
- Issues with algorithms
 - Overfitting
 - Underfitting

Machine Learning Workflow

- Textbook /'idealized' workflow from Chapter 2 - End to end machine learning project
 - Look at the big picture
 - Get the data
 - Discover and visualize the data to gain insights
 - Prepare the data for machine learning algorithms
 - Select a model and train it
 - Fine-tune your model
 - Present your solution
 - Launch, monitor and maintain your system

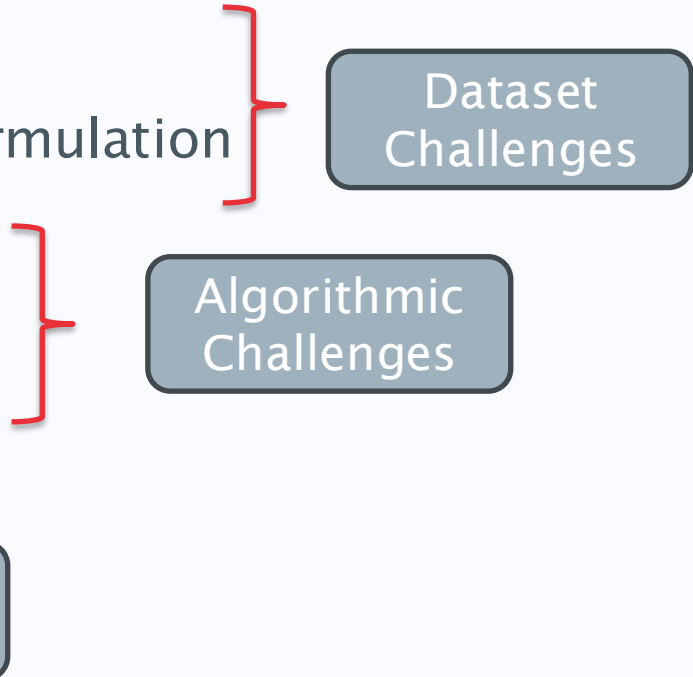


**However,
Machine Learning
Algorithm Design
is an Iterative Process**

Machine Learning in Practice

- Practical iterative workflow

- Problem characterization
- Data characterization
- Data analysis & hypothesis formulation
- Model Selection and Training
- Evaluation setup & analysis
- *Iterative improvement*
- Release
- Maintain →



Dataset Challenges

Algorithmic Challenges

Monitoring performance

Problem Characterization

- Objectives
 - What do you want to accomplish with ML system?
 - E.g. Understanding the house purchasing patterns of individuals
- Success metrics –
 - What results indicate success?
 - What is a good accuracy/speed?
- Customer constraints
 - Computational facilities
 - Timelines
- Flexibility
 - Exploration of new methods or other objectives
- Existing solutions or similar problems?
- Assumptions

Data Characterization - Availability

- Data availability
 - Are there any datasets?
 - Are there some available labels? → Supervised/Unsupervised
 - What attributes/features are available?

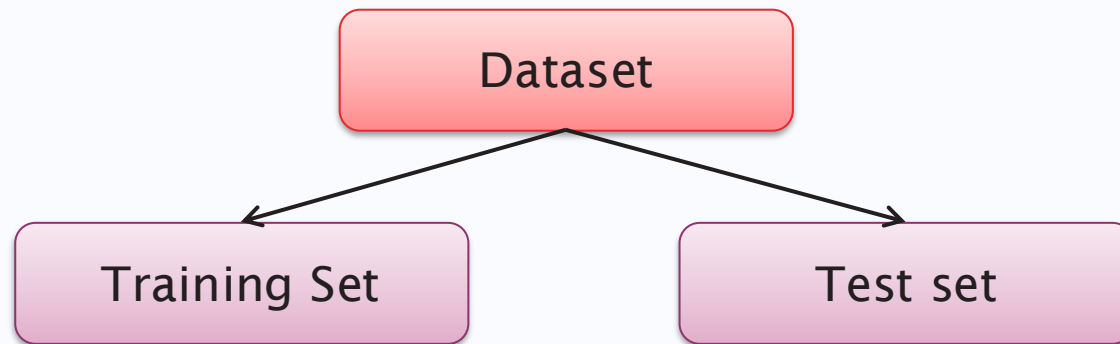
Age	Income	Gender	Education	Marital_Status	Purchased
56	-32695	Male	PhD	Single	Yes
46	78190	Female	High School	Divorced	No
14	35258	Male	PhD	Single	No
60	117538	Male	-	Single	Yes
-	69504	Male	Masters	Divorced	Yes

Snapshot of a dataset

Legal & ethical considerations

- Legal & ethical considerations
 - Legal considerations
 - Is it open data or closed data?
 - Licenses? Any other regulations?
 - Ethical considerations
 - UK Data Ethics Framework – Overarching Principles
 - Transparency
 - Accountability;
 - Fairness
 - Ethical approvals
 - Data collection – experiment design, informed consent, data storage
 - Secondary data – sources, potential biases

Training vs Test Data



- To get a sense of how well the model generalises, you need to test it on unseen instances
- To facilitate this, set aside a test set to avoid *snooping bias*
- You are to train a model on the training data
- You (or someone else) will use this model to process testing data
- Training data and testing data do NOT necessarily share the same properties

Data Analysis – Understanding the data

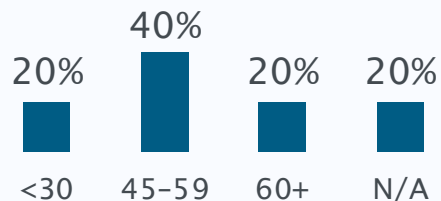
- Identify the data attributes/features?
 - Name
 - Type (categorical, int/float, bounded/unbounded, text, structured, etc.)
 - Usefulness for the task

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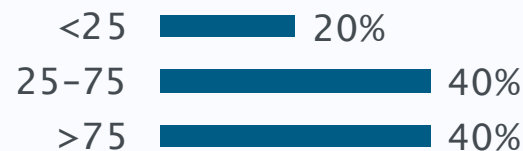
Data Analysis and Hypothesis Formulation

- Visualise the data
 - Range of values
 - Type of distribution (Gaussian, uniform, logarithmic, etc)
- Identify patterns & correlations
- Form hypotheses

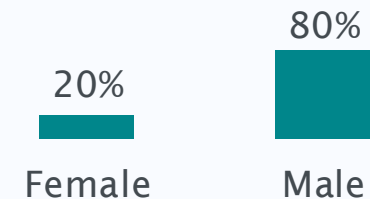
Age



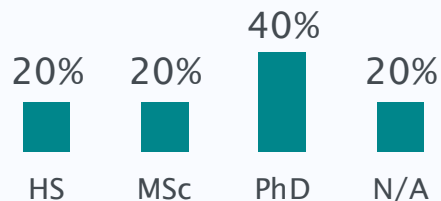
Income (£k)



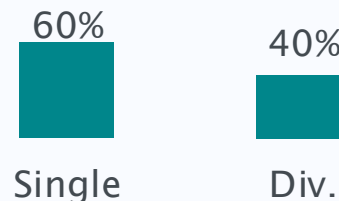
Gender



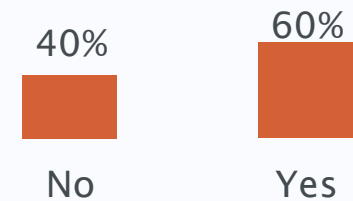
Education



Marital status



Purchased



Dataset Challenges

- Insufficient Training Data
 - ML methods are data-hungry
 - Ranging from ~1 000 data points for simple problems to millions for image or speech recognition
 - Merge datasets? Collect new data?
- Non-representative Data
 - Training set should be representative of the cases we want the model to generalize to
 - If the dataset is too small, we will have *sampling noise* – non-representative as a result of chance
 - Large datasets can still be non-representative. This leads to *sampling bias*. E.g. non-inclusive data collection methods like online surveys can exclude older populations.

Dataset Challenges II

- Poor Quality Data
 - Can you identify the poor quality data points in the table below?
 - Errors, outliers, noise and missing data

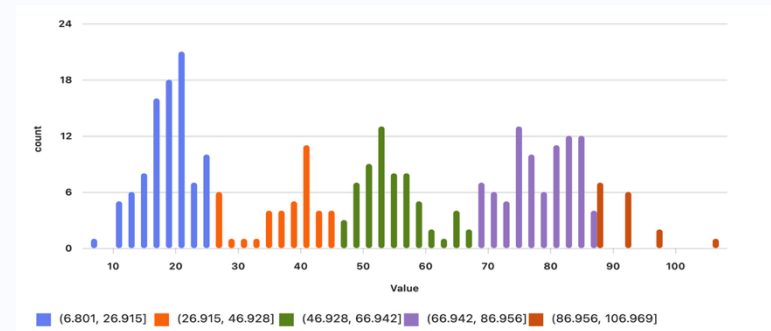
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Dataset Challenges

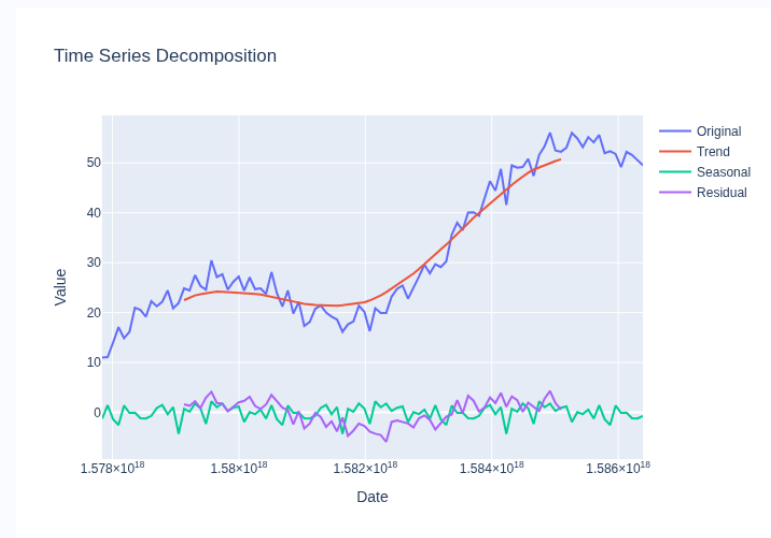
- Poor Quality Data
 - Errors, outliers and noise
 - Missing features/attributes
 - Discard the data instances
 - Fill in the missing values?
 - Train models without these features?
- Irrelevant Features
 - Not all features are important in making a prediction
 - Feature Engineering -
 - Select features – dimensionality reduction
 - Create new features

Data Preparation

- Pre-processing
 - Removing noise
 - Feature engineering
 - Discretization
 - Decomposition
 - Simple, interpretable components
 - Transformation
 - Aggregation
 - Feature scaling
 - Min-max; normalization



Discretization



Decomposition

Model Selection and Training

- No Free Lunch Theorem
 - No model is a *priori* guaranteed to work better
 - You will have to evaluate all the models to know which is the best. Unless... you make some **reasonable assumptions** about the data and the problem.
- Machine learning algorithm selection
 - Algorithm attributes need to suit the problem and data characteristics – E.g. you may use linear models for simpler tasks and neural networks for a complex problem
 - What are the most significant variables for the algorithm?
 - What errors does the algorithm usually make?
 - You can train *quick versions* of different suitable models with standard parameters first
 - Compare the performance and shortlist

Performance Measures

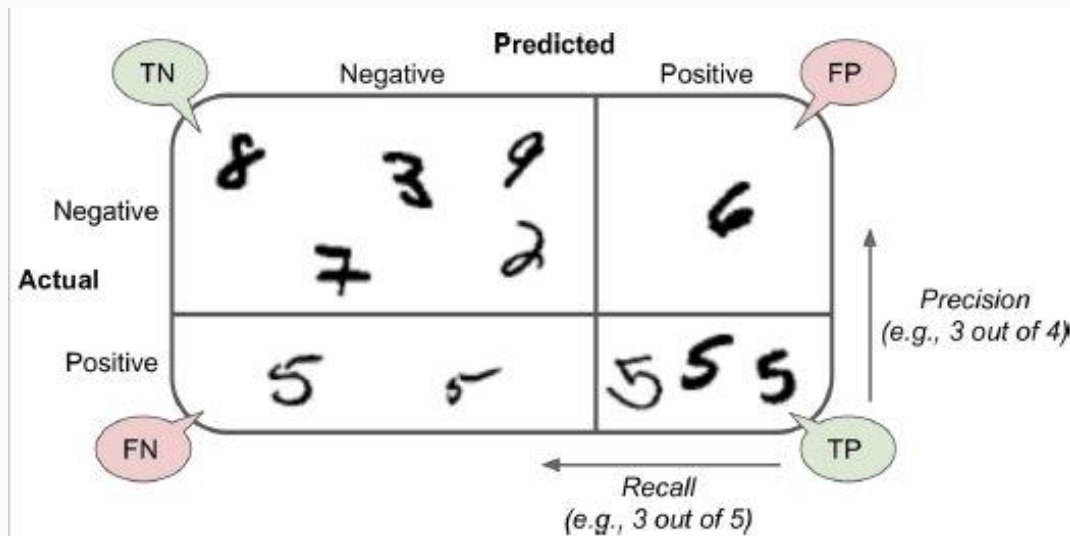
- Goal: measure the goodness of the ML model
- Standard performance metrics:
 - Root Mean Square Error (RMSE) = $\sqrt{\frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2}$
 - Mean Absolute Error (MAE) = $\frac{1}{m} \sum_{i=1}^m |y_i - \hat{y}_i|$
 - The smaller the error, the more accurate is the model
 - RMSE and MAE are typically used in regression models
 - For classification, we use other (simpler) functions
 - A simple function: percentage of correct predictions made by the classifier (we will refer to this as accuracy)

Accuracy as a performance measure

- Example: asthma attack prediction from cough sound
 - 99% of cases: no attack after coughing
 - 1%: asthma attack
- We train a model and get 99% accuracy
 - Is this good enough?
- Think about this: a very dumb algorithm – always says “no”
 - And still is accurate 99% of the time
- Reason?
 - An imbalanced (skewed) dataset
- Solution: A new metric which can capture misclassification separately for each class instead of showing the aggregated accuracy.

Confusion Matrix

- Example: classify whether a hand-written digit is 5 (i.e., a binary classifier)
 - True positive (TP): “Yes” data correctly predicted as “Yes”
 - True negative (TN): “No” data correctly predicted as “No”
 - False positive (FP): “No” data incorrectly predicted as “Yes” (Type I error)
 - False negative (FN): “Yes” data incorrectly predicted as “No” (Type II error)



Precision, Recall and F1-score

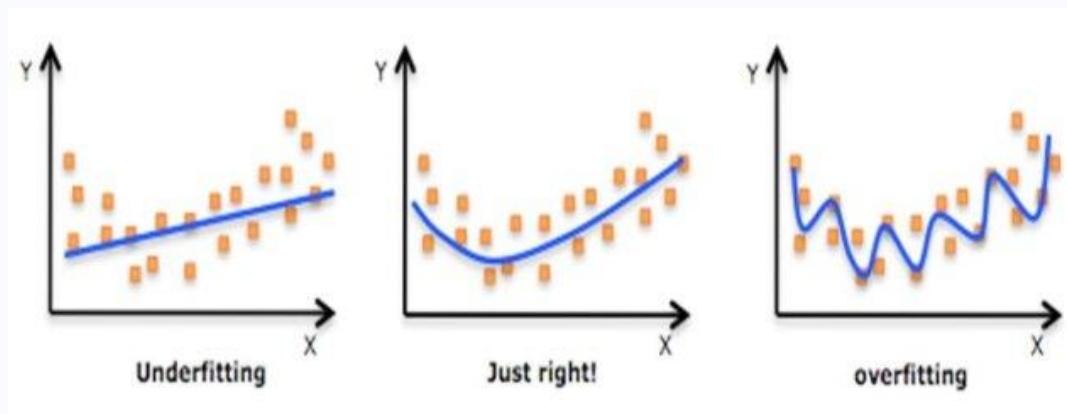
- $Precision = \frac{TP}{TP+FP}$; ratio of correct “Yes” predictions
- $Recall = \frac{TP}{TP+FN}$; ratio of “Yes” correctly predicted
- $F1 = \frac{2}{\frac{1}{precision} + \frac{1}{recall}} = 2 \times \frac{precision \times recall}{precision + recall} = \frac{TP}{TP + \frac{FN+FP}{2}}$
 - F1 captures both precision and recall; harmonic mean of the two values
 - High only if both values are similarly high
- We may sometimes prefer Precision over recall or vice-versa
 - Example: child-friendly video detection: prefer to reject many good videos (low recall) but keeps only safe ones (high precision)

Evaluation Setup and Analysis

- Visualize the results
- Examine the confusion matrix
- Fault analysis
 - Can you identify the examples where the model doesn't work?
- Causal analysis
 - What is causing the resultant performance?
- Statistical Significance
 - Is the algorithm performing *significantly* better than others?

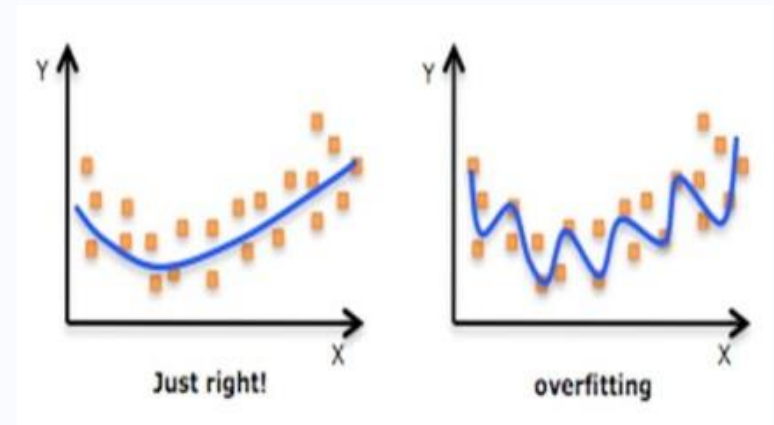
Algorithm Challenges

- Generalization error
 - Error rate on the new cases (generally from the test data set aside)
- Underfitting
 - Low performance on training data
- Overfitting
 - Training error is low, but generalization error is high



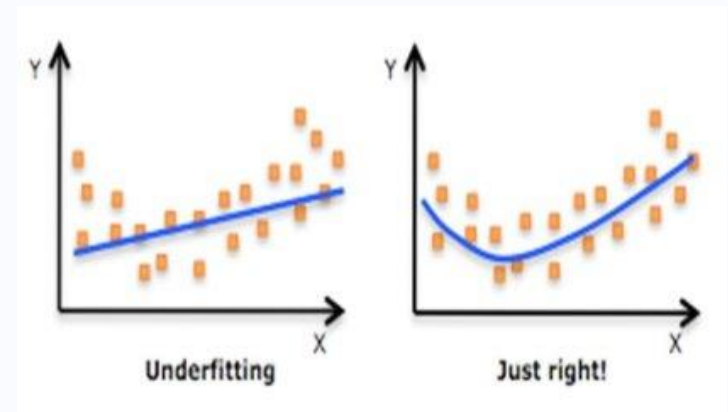
Overfitting

- Mathematically, your model has a higher degree of non-linearity.
- Possible solutions
 - Simplify the model
 - Use one with fewer parameters
 - Or constrain the parameters – *regularization* (to be discussed in the coming lectures)
 - Get more training data – improves the generalization error



Underfitting

- Your model is not complex enough
- Possible solutions
 - Select a more complex model – more parameters; higher non-linearity
 - Reduce constraints on the model parameters – lower the regularization parameter
 - Feed better features – explore feature engineering techniques



Iterative Improvement

- Improve feature selection
 - Experiment with removing or adding features
- Explore parameter space
 - Hyperparameter search and tuning (to be discussed in the coming lectures)
- Get more training data
 - Experiments? Buy new/large datasets?
- Parallelize
- Test alternative algorithm designs

Release Your Solution

- Document all the data and algorithmic choices
- Explain how the objectives are achieved
- Learning from your experience
 - What alternatives have you tried?
 - What worked and what didn't? Why?
 - What assumptions were made?
 - What are the limitations?
 - How can you make it better in the future?

Maintain

- Monitor the system
 - Logs to know when something goes wrong
 - Performance degradation over time
 - Data trends can change – Model drift
 - Concept drift - relationship between input features and the target variable changes. E.g. features of spam email may change over time
 - Data drift (Covariate shift) - distribution of input features changes, but the relationship to the target remains the same. E.g. shift in customer demographics
 - Label drift - distribution of the target variable changes. E.g. sudden increase in positive cases during a pandemic
 - May need to retrain from time to time
 - Human oversight may be needed
 - Create a meaningful interface for the users to interact

Summary

- In this lecture, we looked at a practical end-to-end workflow for developing a machine learning system
- Detailed steps/checklist for creating a machine learning project
- Identified challenges and possible solutions at each stage
- This framework should be adapted based on your needs
- We take a deeper look into these steps as we learn about different ML models in detail.
- This checklist will also be useful while working on your coursework

Next Lecture: Regression Models

Questions?